Wind Speed:

Field	Definition
Parameter	Wind Speed
Category	Environment
Unit	m / s
Role	Continuous model input / predictive feature
Controllable?	No
Measured or Derived?	Measured
Expected Range	Proposed modeling range: 0-7 m/s (~0-15.7 mph). Actual allowable application range is label dependent. EPA guidance commonly used wind-speed limits in the 10-15 mph range depending on the application and label.
Primary Relationships	Increasing wind speed generally increases horizontal droplet displacement and drift potential. Its effect depends strongly on droplet size/ settling velocity, release height, and wind direction.
Output Affected	Primarily off-target deposition/drift; secondarily target deposition, coverage, and coverage uniformity.
Source/ Data Needed	On-machine or field anemometer; synchronized wind measurement during spraying. Wind should ideally be measured at or above spray release height and away from major obstructions.

Measurement Definition:

Wind speed should represent ambient air velocity near the spray release height during application, rather than distant weather-station measurement. EPA guidance states that wind speed used for spray-drift decisions should be measured at release height or higher and in an area free from obstructions such as trees, buildings, and equipment.

Simplified Physical Relationship

The first-order relationship we decided to use is:

$$x \cong U\ (H/Vt)$$

=

where:

- x = estimated horizontal droplet displacement, m
- U = horizontal wind speed, m/s
- H = droplet release height above the target, m
- Vt = terminal settling velocity of the droplet, m/s

This comes from combing:

$$x\ =\ Ut$$

with:

$$t = H / Vt$$

therefore:

$$x = U\,(H / Vt)$$

Interpretation:

This relationship tells us:

$$U \uparrow \Rightarrow x \uparrow$$

Higher wind speed increases horizontal displacement

$$H \uparrow \Rightarrow x \uparrow$$

Greater release height gives the droplet more time to move horizontally.

$$Vt \uparrow \Rightarrow x \downarrow$$

Droplets with greater settling velocity reach the target faster and therefore have less time to drift.

Droplet settling velocity reference

Rather than calculating terminal velocity ourselves, let's use published droplet settling velocities.

Table II.7–1—Selected characteristics of various size spray droplets of water

Droplet diameter	Terminal velocity	Fall time from 3 m	Drift distance (3-m fall with 5-km/h wind)	Drops/cm² from 10 a/ha application
(μm)	*(M/sec)*	*(Sec)*	*(M)*	*(No./cm²)*
10	0.003	1,020	1,372	190,990
50	0.075	40	54	1,530
100	0.279	11	15	192
200	0.721	5.4	5	24
500	2.139	1.6	2	1.5

The same USDA table also demonstrates the relationship between droplet diameter, fall time, wind speed, and drift distance, making it a useful supporting source for the simplified displacement approach.

Assumptions of this relationship

The equation

$$x \cong U\,(H/Vt)$$

should be described as a first-order or simplified drift-distance approximation, not as the project's final drift model.

It assumes:

- Constant wind speed
- Constant droplet settling velocity
- Approximately horizontal wind
- No evaporation
- No turbulence
- No changing droplet diameter

- No canopy interception
- No wake effects from the sprayer
- No detailed nozzle exit velocity effects

So the purpose of this equation is to show the expected physical relationship between variables, not to perfectly predict real-world drift.

Conclusion: Wind speed is a continuous environmental input measured in m/s near spray release height. Increasing wind speed generally increases off-target droplet displacement, with the magnitude of the effect strongly influenced by release height and droplet settling velocity. A first-order displacement relationship is $x \cong UH / Vt$, which may be used as a physics-informed feature or conceptual baseline rather than as the final drift model.

Sources:
https://www.epa.gov/pesticides/mitigation-menu-measure-descriptions
https://www.ars.usda.gov/ARSUserFiles/30320505/grasshopper/Extras/PDFs/IPM%20Handbook/SectionII-Chemical.pdf
https://www.epa.gov/pesticide-registration/prn-2001-x-draft-spray-and-dust-drift-label-statements-pesticide-products

Wind Direction:

Field	Definition
Parameter	Wind Direction
Category	Environment
Unit	Degrees (°), preferably converted to a relative angle or vector components for modeling
Role	Continuous environmental input / directional predictive feature
Controllable?	No
Measured or Derived?	Measured, then optionally derived into relative-angle or vector components.
Expected Range	0° - 360°
Primary Relationships	Determines the direction in which spray droplets are transported; interacts directly with wind speed, sprayer travel direction, field orientation, and location of off-target/ sensitive areas
Output Affected	Primarily direction and spatial distribution of off-target deposition/ drift; also target deposition and coverage distribution
Source/ Data Needed	Wind-direction sensor plus vehicle heading/GPS

Measurement Definition:
For this project, we will assume the sensor/preprocessing layer provides the direction the airflow is moving toward, so the model does not need a separate meteorological “from-direction” conversion

$$\theta w \;=\; airflow\ direction$$

with:

- 90° = wind from east
- 180°= wind from south
- 270° = wind from west
- 360° = wind from north

Relative Wind Direction:
Absolute compass direction is less useful to the model than the direction of airflow relative to the machine’s travel direction.

Let:

$$\theta w \;=\; wind\ angle$$

and

$$\theta v \;=\; vehicle\ heading$$

Then we define the relative angle as:

$$\theta rel = \theta w - \theta v$$

The results can then be normalized to a range such as:

$$-180° \leq \theta rel \leq 180°$$

EPA spray-drift field guidance supports the importance of wind direction relative to machine travel. EPA specifies that deposition collectors should be arranged appropriately parallel to the wind direction and as close as possible to perpendicular to the direction of application equipment travel.

Wind-vector Components:
Once wind speed U and relative direction θrel are known, the wind can be separated into:

- A component parallel to machine travel
- A component perpendicular to machine travel

If U is wind speed and θrel is the angle relative to the vehicle path, then:

$$Ualong = Ucos(\theta rel)$$

and

$$Ucross = Usin(\theta rel)$$

Formulas derived from standard vector-component formulas with the vehicle travel direction chosen as one axis.

Interpretation:

If:

$$\theta rel = 0°$$

Then:

$$Ucross = 0$$

And all wind is parallel to travel.

If:

$$\theta rel = 90°$$

Then:

$$Ucross = U$$

And wind is entirely perpendicular to travel.

Connection to the simplified drift relationship

For wind speed, the first-order relationship is:

$$x \cong U\ (H/Vt)$$

For cross-track displacement, it is more meaningful to use only crosswind component:

$$x\ cross \cong Ucross(H/Vt)$$

Substituting:

$$Ucross = Usin(\theta rel)$$

Gives:

$$x\,cross \cong Usin(\theta rel)(H/Vt)$$

This is a project-derived first-order approximation, not EPA or published pesticide-drift equation. EPA's role here is supporting the physical relevance of wind orientation relative to machine travel and downwind deposition, not supplying this exact equation.

Recommended model treatment

For the raw dataset, retain:

Wind Direction
Wind Speed
Vehicle Heading

Then derive:

θrel
$Ucross$
$Ualong$

The derived components are likely more useful to the model than raw compass direction alone because they describe airflow relative to the spraying operation.

Conclusion: Wind direction is a continuous environmental input representing the direction in which ambient airflow is moving. For this project, any conversion from meteorological "from" direction is assumed to occur at the sensor or preprocessing stage. Wind direction is compared with vehicle heading to calculate relative wind angle. Standard vector decomposition is then used to determine crosswind and along-track wind components, which influence direction and spatial distribution of spray deposition and drift.

Sources:
https://forecast.weather.gov/glossary.php?word=DIR
Spray Drift Test Guidelines OPPTS 840.1200 Spray Drift Field Deposition

Temperature & Relative Humidity:

Field	Air Temperature	Relative Humidity
Category	Environment	Environment
Unit	°C	%
Role	Continuous model input	Continuous model input
Controllable?	No	No
Measured or Derived?	Measured	Measured
Expected Range	10 - 40°C	10 - 100 % RH
Primary Relationships	Higher temperature generally increases droplet evaporation, especially when relative humidity is low	Lower RH generally increases droplet evaporation, especially when temperature is high
Output Affected	Off-target deposition/drift, target deposition, coverage	Off-target deposition/drift, target deposition, coverage
Source/ Data Needed	Ambient temperature sensor	Relative-humidity sensor

Why they are grouped together:

Temperature and relative humidity are most useful in this project because they jointly influence spray-droplet evaporation. University of Nebraska Extension states that low relative humidity and/or high temperature cause faster evaporation of droplets and increasing drift potential. It also notes that hot, dry conditions can reduce deposition on the target.

Key relationship for the model is:

$$T\uparrow,\ RH\downarrow \Rightarrow evaporation\ rate\uparrow$$

Followed by:

$$evaporation\uparrow \Rightarrow d\downarrow$$

Where d is droplet diameter.

Connection to drift

As a droplet evaporates, its diameter decreases. Smaller droplets are more easily transported by ambient air currents and generally remain suspended longer.

So our conceptual chain is:

$$T\uparrow,\ RH\downarrow \rightarrow evaporation\uparrow \rightarrow d\downarrow \rightarrow Vt\downarrow \rightarrow drift\ susceptibility\uparrow$$

This ties directly back to the wind-speed relationship we already established:

$$x \cong U\ (H/Vt)$$

If evaporation reduces settling velocity Vt, then the droplet remains airborne longer and has more time to move horizontally. This should be treated as a physical relationship and model rationale, not a complete evaporation model.

Experimental Support:

Zhu et al. simulated spray-droplet drift using:

$$10℃ \leqq T \leqq 30℃$$

And:

$$10\% \leqq RH \leqq 100\%$$

along with wind velocity, droplet size, release height, initial droplet velocity, and turbulence. They found that temperature and humidity had a much greater influence on the drift distance of droplets smaller than 100 um than on droplets 200 um and larger. Their simulations also found that, except under low-temperature/high humidity conditions, droplets around 50 um and smaller could completely evaporate before depositing over relatively short vertical distances.

This means the interaction is:

$$T \; x \; RH \; x \; d$$

Droplet-size sensitivity
University of Florida IFAS gives a useful practical breakdown:

- Droplets ≦ 50 um may evaporate very rapidly before reaching the target
- Droplets ＞200 um generally experience much less size reduction before deposition
- Droplets roughly 50-200 um are particularly influenced by temperature, humidity, and other weather conditions

Conclusion: Air temperature and relative humidity are separate continuous environmental inputs documented together because of their combined influence on droplet evaporation. High temperature and low relative humidity increases evaporation, reducing droplet diameter and increasing susceptibility to off-target transport. Their effect is strongly dependent on droplet size, with smaller droplets being substantially more sensitive than large droplets. Temperature and relative humidity therefore primarily influence off-target deposition/ drift, target deposition, and coverage through their effect on droplet evaporation.

Sources:
https://extensionpubs.unl.edu/publication/g1773/na/html/view
https://elibrary.asabe.org/abstract.asp?aid=28220&

Travel Speed:

Field	Definition
Parameter	Travel Speed / Ground Speed
Category	Machine / Operational
Unit	Km/h or m/s
Role	Continuous model input and controllable operating parameter
Controllable?	Yes
Measured or Derived?	Measured
Expected Range	Use the actual operating range of the machine
Primary Relationships	Directly interacts with nozzle flow rate, spray width/nozzle spacing, and application rate. If flow is held constant, increasing travel speed decreases spray volume applied per unit area.
Output Affected	Application rate, target deposition, spray coverage, coverage uniformity; potentially off-target deposition indirectly
Source/ Data Needed	Machine CAN bus, wheel-speed sensor, GPS/GNSS, or other synchronized machine-speed measurement

ASABE basis:

The primary standard for this parameter is ASAE EP367.2 – Guide for Preparing Field Sprayer Calibration Procedures. ASABE states that the Engineering Practice provides guidance for calibrating boom-type field sprayers used for broadcast, band, or row applications, with the goal of improving the uniformity, accuracy, and safety of pesticide application. Travel speed is a fundamental calibration variable because it determines how much ground area is covered during a given period of spray flow.

Critical application-rate relationship

For a broadcast boom sprayer in SI units:

$$q = (ARws) / 600$$

where:

- q = total boom flow rate, L/min
- AR = application rate, L/ha
- w = treated boom width, m
- s = travel speed, km/h

Rearranged to solve for application rate gives:

$$AR = (600q) / ws$$

Primary travel-speed relationship:

If boom width and flow remain constant:

$$AR = (600q) / ws$$

therefore:

$$AR \propto 1 / s$$

So:

$$s \uparrow \Rightarrow AR \downarrow$$

Only when q and w remain constant

Rate-control interaction

A modern closed-loop or rate-controlled sprayer can respond to increased speed by increasing flow:

$$s \uparrow \Rightarrow q \uparrow$$

Such that:

$$AR = constant$$

From the ASABE relationship:

$$q = (ARws) / 600$$

If AR and w are fixed:

$$q \propto s \text{ when } AR \text{ and } w \text{ are constant}$$

So if travel speed rises by 20%, the required total flow must also rise by approximately 20% to maintain the same application rate.

Connection to deposition and coverage

Relationship:

$$s \rightarrow AR \rightarrow deposition/coverage$$

So:

Without rate compensation

$$s \uparrow \Rightarrow AR \downarrow \Rightarrow less\ spray\ volume\ per\ unit\ area$$

With rate compensation

$$s \uparrow \Rightarrow q \uparrow \Rightarrow AR \cong constant$$

Recommended Model Treatment

Travel speed should remain a continuous variable:

$$s = ground\ speed\ in\ km/h$$

The primary interactions to retain are:

$$s\ x\ q$$

Travel speed and total flow rate,
and:

$$s\ x\ w$$

travel speed and width.

For the eventual closed-loop controller, travel speed could also be one of those variables the system adjusts in response to predicted coverage or deposition performance.

Conclusion: Travel speed is a continuous and controllable machine parameter representing the forward ground velocity of the sprayer. ASAE367.2 establishes field sprayer calibration as a process involving interacting application variables, and ASABE engineering references express application flow using $q = ARws/600$. Consequently, at constant boom width and flow rate, application rate is inversely proportional to the travel speed to maintain a constant application rate. Travel speed should therefore be modeled jointly with flow rate and spray width rather than treated as an independent predictor of deposition or coverage.

Sources:
https://elibrary.asabe.org/abstract.asp?aid=24435&redir=&redirType=&t=28

Boom Height:

Field	Definition
Parameter	Boom height / spray release height
Category	Machine / Spray system
Unit	m
Role	Continuous model input and potentially controllable operating parameter
Controllable?	Yes, if the machine has active boom-height control
Measured or Derived?	Measured
Expected Range	Equipment and nozzle dependent. 0.3-0.75 m above target/canopy based on published ASABE field testing.
Primary Relationships	Increasing boom height increases droplet airborne time and therefore potential wind displacement; boom height also affects spray-pattern overlap and coverage uniformity
Output Affected	Off-target deposition/drift, target deposition, coverage, coverage uniformity
Source/ Data Needed	Boom-height sensor, ultrasonic/radar sensor, machine CAN data, or known commanded boom height.

ASABE field research by Nuyttens et al. evaluated boom heights of 0.30, 0.50, and 0.75 m and found that lower boom heights generally reduced spray drift. Their reference configuration used a 0.50m boom height and 0.50m nozzle spacing

Why boom height matters:
Boom height affects the amount of time spray droplets spend suspended between the nozzle and the target. The EPA applicator manual states directly that the higher the boom is above the target, the greater the potential for drift, and that lowering the boom can reduce off-target movement.

EPA also currently distinguishes between:

$$Low\ boom \leq 24\ in$$

and

$$High\ boom > 24\ in$$

Above ground or crop canopy for ground-boom drift mitigation. For fine-droplet applications, lowering the boom to 2 ft or less can qualify for substantial drift buffer reduction. However, for this model, we should retain continuous height.

Main physical relationship

If a droplet falls a vertical distance H at settling velocity Vt, then its approximate airport time is:

$$t = H / Vt$$

Where:

- t = approximate airborne time, s
- H = boom/release height above target, m
- Vt = droplet settling velocity, m/s

This means:

$$H \uparrow \Rightarrow t \uparrow$$

A higher boom gives the droplet more time in the air.
Then horizontal displacement due to wind is approximated by:

$$x = Ut$$

Substituting the fall-time relationship:

$$x \cong U\,(H / Vt)$$

where:

- x = horizontal droplet displacement, m
- U = wind speed, m/s

Therefore, if wind speed and droplet settling velocity stay constant

$$x \propto H$$

Simplified in this model:

$$H \uparrow \Rightarrow xdrift \uparrow$$

So increasing height by 50% produces approximately 50% more displacement under the assumptions of this simple model.

Spray Pattern Overlap
Boom height has another important effect besides drift: it changes how adjacent nozzle spray patterns overlap at the target. Flat-fan nozzles create expanding spray patterns as distance from the nozzle increases. Boom height, nozzle spacing, and spray angle therefore work together to determine overlap.

Conceptually:

$$Boom\ Height + Nozzle\ Angle + Nozzle\ Spacing \rightarrow Pattern\ Overlap$$

This matters because inadequate overlap can cause streaking, underapplication between patterns, and poor coverage uniformity. So there is a tradeoff, if boom is to low then potential insufficient overlap while boom to high can lead to greater drift potential.

The model should receive boom height, nozzle type/ spray angle, and nozzle spacing then learn their relationship to coverage.

Optional spray geometry
Here is a simple geometric relationship for the data book where a spray fan with included angle a at height H has an approximate theoretical width:

$$Ws = 2H\ tan\,(\alpha/2)$$

where:

- Ws = theoretical spray pattern width
- H = nozzle height above target
- a = spray angle

So:

$$H \uparrow \Rightarrow Ws \uparrow$$

For a fixed spray angle

Boom height x nozzle spacing
Important practical relationship is:

$$Pattern\ overlap\ depends\ on\ (Ws\ /\ S)$$

where:

- Ws = spray fan width at target
- S = nozzle spacing

Interaction with wind and droplet size
Boom height should not be modeled independently.

Its drift effect depends heavily on:

$$H\ x\ U$$

Boom height x wind speed
And:

$$H\ x\ Vt$$

Boom height x droplet settling velocity

The combined first-order relationship is still

$$x \cong (UH)\ /\ Vt$$

This tells the model that raising the boom is especially problematic when wind speed is high and droplets are small or have a low settling velocity.

Recommended model treatment
Store boom height as:

$$H\ =\ continuous\ height\ above\ target/canopy\ in\ meters$$

Conclusion: Boom height is a continuous and potentially controllable spray-system parameter representing the vertical release height of the nozzle above the target or crop canopy. Increasing boom height increases droplet airborne time, increasing the opportunity for wind-induced displacement and off-target deposition. ASABE field research evaluating boom heights of 0.30, 0.50, and 0.75 m found that lower boom heights generally reduced spray drift. Boom height also interacts with nozzle spray angle and spacing to determine spray-pattern overlap and coverage uniformity. The parameter should therefore be modeled jointly with windspeed, droplet size/ settling velocity, nozzle characteristics, and target height.

Sources:
https://elibrary.asabe.org/abstract.asp?aid=23622&utm
Pesticide Applicator Certification Core Manual Second Ediditon